Tolerance to Deviancy in Adolescents

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Introduction

Data showing the tolerance to deviant behaviour of youths (given by an average of a Likert score of how they feel about different types of behavious) are provided. Each youth was asked the same questions each year, and we have a score for exposure to deviant behaviour and their gender. I have included the code used to produce the report, however I have written explanations in the text, so if you don't understand the code, you should still be able to understand the report.

First, we load the libraries and the data to do the analysis and make a new variable, `CAge` is the time they are in the study (so an 11 year old is now listed as 0 and a 15 year old is now listed as 4). This doesn't change the results of the analysis, it just makes it slightly easier to interpret as `Cage` is the number of years since they were 11.

```
library(lme4)
```

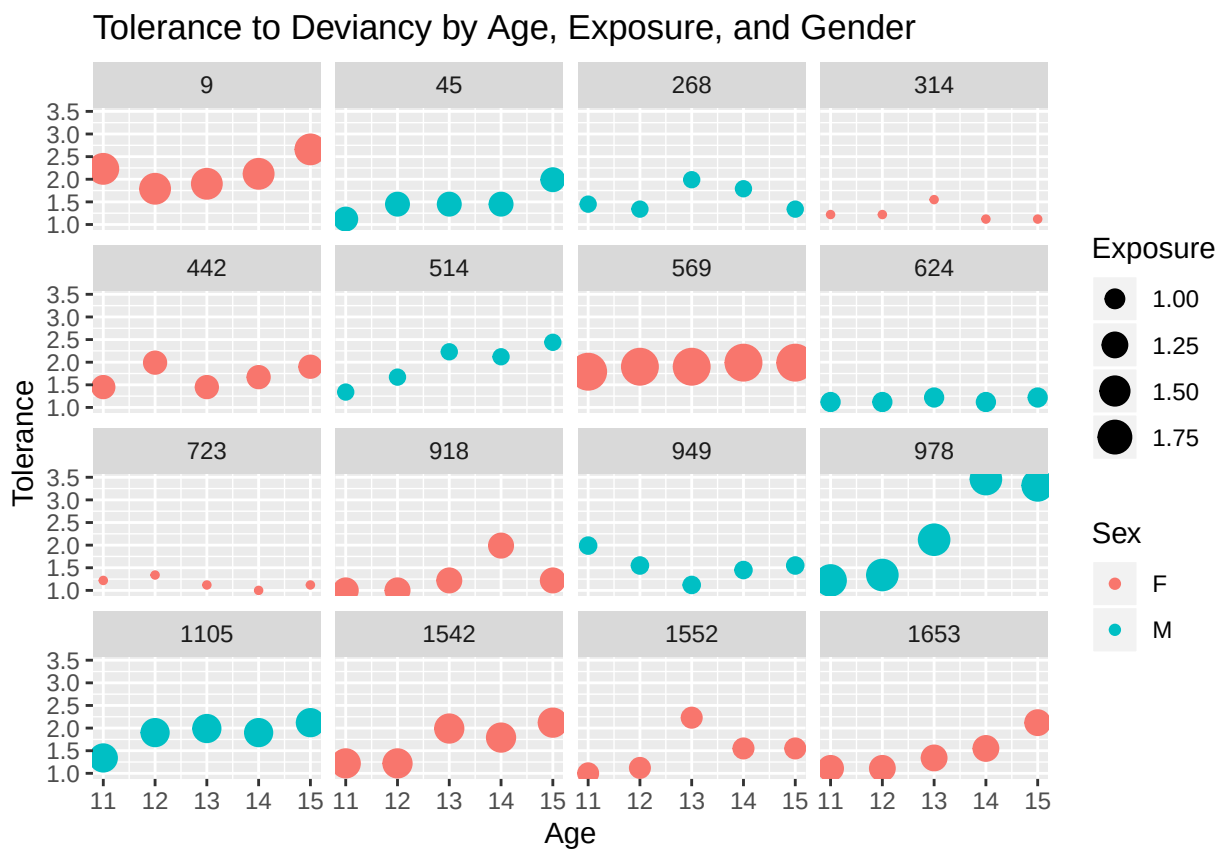
```
## Loading required package: Matrix
```

```
library(ggplot2)
library(gridBase)
library(mvtnorm)
load("/mnt/udrive/Portfolio/Tolerance to Deviancy/Tolerance Data.RData")
YouthSurvey$CAge <- YouthSurvey$Age - 11
```

Exploratory Data Analysis

An exploratory plot of the data, showing tolerance by age for each of the 16 youths in the study is below. I have also included the Sex as a colour in each plot, blue represents male and red represents female.

```
  qplot(x = Age,
        y = Tolerance,
        facets = ~id,
        col = Sex,
        size = Exposure,
        data = YouthSurvey,
        main = "Tolerance to Deviancy by Age, Exposure, and Gender")
```



These graphs have tolerance increasing with age in almost all cases, which is what most people would expect to see. There is some variation in how much it increases within each youth and also in the tolerance at age 11, suggesting a random effect is required. It is hard to tell whether sex and exposure make a difference, so I expect that age will be the most important factor.

Fitting a Full Model

I have fit a 'full' model based on what I think is plausible given the graphs above. This compares tolerance to age, sex, and exposure, with a random effect for the age affecting the individual. The `lmer` function fits a mixed effects model with the following features:

- `Tolerance` is my response
- `Sex + Exposure + CAge` describe the fixed effects. I could have used `Sex * CAge` instead of `+`, and this would have given an interaction, where males have a different trend line to females, however I don't think I have enough data or a need to fit this.
- The `(CAge|id)` describes the random effect, this tells the computer that I want to fit a model with a straight line of different gradient and intercept for each person. If my data had more nesting levels (for instance each person was in a different school) I could fit this with `(CAge | id:school)`. If I typed `(1 | id)`, this would have fit an intercept only, with a gradient that is constant across all youths.
- `REML = F` means that I will fit the model with reduced likelihood, this allows me to perform hypothesis tests later.

```
model_full <- lmer(Tolerance ~ Sex + Exposure + CAge + (CAge | id),
                   data = YouthSurvey,
                   REML = F)
```

Hypothesis testing

I will test whether the random effect for Age is necessary in the model. My H_0 (null hypothesis, or what I will be assuming is true unless I find evidence against) is that $b = 0$ and my H_A (alternative hypothesis) is that $b \neq 0$.

Some things to note about testing mixed effects models:

- We should test all random effects first before testing fixed effects.
- Adding `REML = F` is necessary because I am using the generalised likelihood ratio test and must therefore use maximum likelihood rather than restricted maximum likelihood.

To run a hypothesis test on the fixed effects, we use a GLRT (Generalised Likelihood Ratio Test): * Fit the full model and the reduced model. * Calculate the observed test statistic for the GLRT . * Calculate the p-value in `R` the function `1 - pchisq(test_stat, df)`. `df` is the number of variables that we have reduced the model by and `test_stat` is the observed test statistic.

To run a hypothesis test on the fixed effects, we use a process called bootstrapping:

- Fit the full model and the reduced model.
- Calculate the observed test statistic for the GLRT .
- For each simulation (more is obviously better):
- Simulate data for the Tolerance from the reduced model
- Fit the full and reduced model on this simulated data
- Calculate the new simulated test statistic
- The approximate p-value is the proportion of simulated statistics which are greater than or equal to the observed statistic from the real data (if we have a large number of simulations but all the simulated test statistics are smaller than the observed one there is probably no point calculating a more precise p-value).
- Because the p-value is approximate and we have N simulations, the smallest p-value we can detect is 1/N. In this case, I will use $N = 100$ for all tests. A higher value of N will give me better results (with diminishing returns), but will also take more time to calculate.

The p-value is the probability that we would see a result at least this extreme if the null hypothesis were true. If the p-value is below 0.05, we will find that there is evidence against the null hypothesis and use the larger model, with smaller p-values representing stronger evidence against the null.

Answering Some Questions about the Data

Now we have gone through what mixed effects models are and how to fit, check, and compare them, we can start to use the data to answer some questions.

Do I need the random effect?

The graphs make this quite clear, but it is always good to check as the human eye is desperate to see patterns that are not necessarily there.

```
# Fit the reduced model
model_reduced_random_effect <- lmer(Tolerance ~ Sex + Exposure + CAge + (1 | id),
                                    data = YouthSurvey,
                                    REML = F)

test_stat_random <- -2*(logLik(model_full) - logLik(model_reduced_random_effect))
attributes(test_stat_random) <- NULL # Get rid of the information we don't need
(p_val_random <- 1-pchisq(test_stat_random, 1))
```

```
## [1] 1
```

A p-value of `round(p_val_random, 5)` is very strong evidence that we do need the random effect to include the gradient. Now we have establised this, we can start to look at some `research` questions that a client might actually want to see.

Does Tolerance Increase with Age?

The graphs looked quite clear that tolerance increases with age, but to check, I am going to fit a reduced model with age missing and perform a bootstrap test.

```
# Fit the reduced model
model_no_age <- lmer(Tolerance ~ Sex + Exposure + (CAge | id),
                     data = YouthSurvey,
                     REML = F)

# Set up boostrap results
bootstrap_results <- rep(0, 10^3)

# Fit bootstrap
for(i in 1:10^3)
{
    # Generate new data from reduced model
    new_data <- YouthSurvey
    new_data$Tolerance <- unlist(simulate(model_no_age))

    # Fit the models to the new data
    new_model_full <- lmer(Tolerance ~ Sex + Exposure + CAge + (CAge|id),
                           data = new_data,
                           REML = FALSE)
    new_model_reduced <- lmer(Tolerance ~ Sex + Exposure + (CAge|id),
                           data = new_data,
                           REML = FALSE)

    # Calculate the test statistic
    bootstrap_results[i] <- -2*(logLik(new_model_reduced) - logLik(new_model_full))
}
p_value_age <- mean(bootstrap_results > -2*(logLik(model_no_age) - logLik(model_full)))
```

A p-value of 0.013 gives us strong evidence against the null hypothesis, that is to say, we need to include the fixed effect term of age.

Does Sex Affect Tolerance?

The graphs do not look very clear on this. Because the graphs are so similar, I am dubious as to whether there is a difference, but I will run a test to find out. For brevity, I won't include the code used, but it is similar to that used for Age and I will report the p-value.

A p-value of 0.208 gives us no evidence against the null hypothesis, that is to say, we do not need to include the fixed effect term of sex because boys have the same tolerance as girls.

Does Exposure Affect Tolerance?

As before, I am dubious as to whether exposure affects tolerance but will perform a test anyway. Again, I will not include the code I have used.

This is very borderline. The p-value of 0.04 is slightly below 0.05, so we have some evidence that exposure affects the tolerance level, however it is very close to the threshold of $p = 0.05$. I will keep it in the model,

but I am also aware that a future study could contradict this.

Checking model assumptions

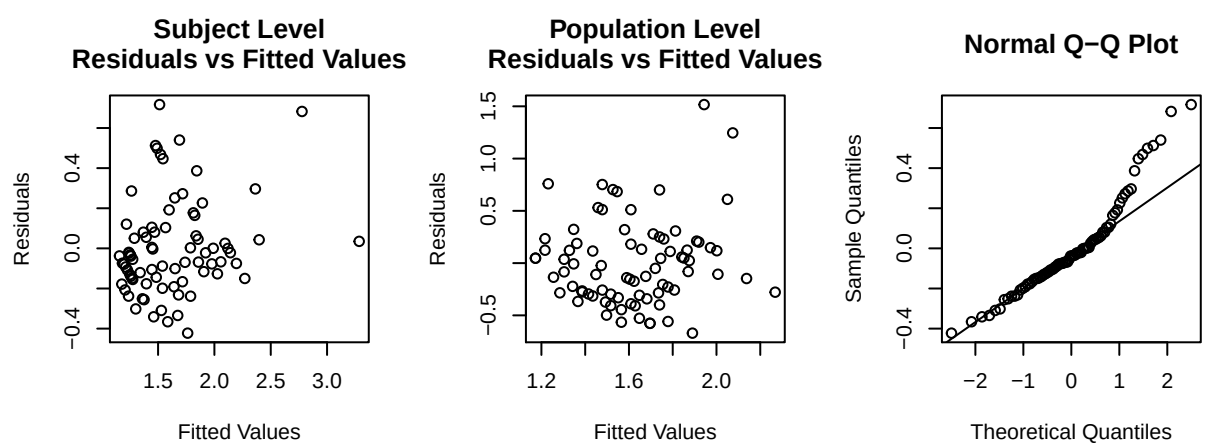
As in any statistical model, we need to check the assumptions. In a mixed effect model, these are that the random effects b_i are i.i.d. with $N(0, \sigma_b^2)$ and that the within-group errors ϵ_{ij} are i.i.d. with $N(0, \sigma^2)$. These can be shown with a plot of:

- Subject-level residuals vs fitted values (we want to see a random scatter)
- Population-level residuals vs fitted values (we want to see a random scatter)
- Normal QQ plot of the subject-wise residuals (we want to see a straight line)

For the final model specified above, these plots are generated below in the order above:

```
par(mfrow = c(1,3))
# Subject level residuals vs fitted values
plot(fitted(model_full), residuals(model_full),
     main = "Subject Level\nResiduals vs Fitted Values",
     xlab = "Fitted Values",
     ylab = "Residuals")
# Population level residuals vs fitted values
# The values for this need to be calculated manually and then plotted
# Multiply the design matrix by the estimated fixed effects to get the fitted values
fitted_level0 <- model_full@pp$X %*% fixef(model_full)
# Calculate the difference between the actual Tolerance and the fitted values
resid_level0 <- YouthSurvey$Tolerance - fitted_level0
plot(fitted_level0, resid_level0,
  main = "Population Level\nResiduals vs Fitted Values",
  xlab = "Fitted Values",
  ylab = "Residuals")

# QQ Plot
qqnorm(resid(model_full))
qqline(resid(model_full))
```



The first two graphs have a random scatter and the third has a mostly straight line, although I will underestimate the higher tolerance values. Therefore, I assume that my assumptions are met and I will use this model for predictions. If this were broken, I would have a few options:

- Use 'robust' methods to re-fit the model in a way that allows for outliers. There are two tradeoffs here, in that I would need a larger sample size for this to give good estimates and try different robust

methods.

- Adjust the parameters, for instance by using a log transformation or taking a quadratic term of one of the predictors. The biggest tradeoff to this adjustment is that it is much less interpretable and this would need to be balanced against the benefit of a more accurate prediction.
- Use a generalised mixed effects model to make sure the predictions are within the range $[1, 5]$ to match the range of the available answers. I don't think this is a big concern here unless we try to extrapolate this model to predict the tolerance of a 10 or 16 year old, which is generally a bad idea.

Predicting a value

The most suitable model available to us is the model found after removing the effect for sex (this is now called `model_full`), and we can use this for predicting future values. As an example, I will use a male who reported an exposure of 1.6. As the final model was dependent on age and exposure only, the gender is not specified in the new data.

```
new_data <- expand.grid(CAge = c(0:4),
                        Exposure = 1.6,
                        id = -1
                        )

preds <- predict(model_full, new_data, allow.new.levels = TRUE)
```

These give the following estimates:

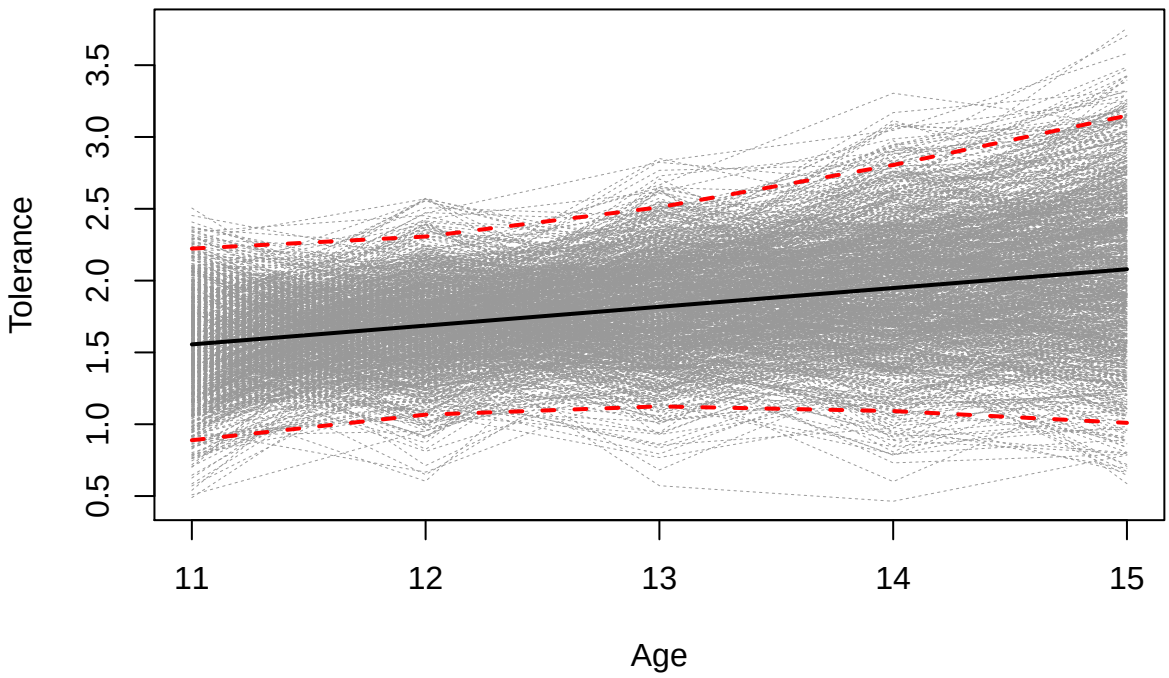
Exposure	11 Years	12 Years	13 Years	14 Years	15 Years
1.6	1.56	1.69	1.82	1.95	2.08

To turn these into a prediction interval to describe some of the uncertainty, I will need to run some more simulations. I need to get a prediction matrix, the values of which are calculated from the numbers from the random effects in the summary of the model. I then simulate new values of these from the multivariate normal distribution. This matrix isn't informative in itself, so I will show the mean prediction (black), randomly generated samples (grey) and 95% prediction intervals (red).

```
rho <- -0.62 # Random effects correlation of CAge
sigma0 <- 0.2044 # Std Dev of intercept
sigma1 <- 0.1429 # Std Dev of CAge
sigma <- 0.2722 # Std Dev of Residual

# Find the prediction matrix
row <- 0:4
sigma_pred <- sigma0^2 +
  sigma1^2 * rbind(row*0, row, row*2, row*3, row*4) +
  rho * sigma0 * sigma1 * rbind(row, row+1, row+2, row+3, row+4) +
  sigma^2 * diag(5)

output <-rmvnorm(10^3, preds, sigma_pred)
matplot(11:15, t(output), type = 'l', lwd = 0.4, col = grey(0.6), lty = 2,
        xlab = "Age", ylab = "Tolerance")
lines(11:15, preds, lwd = 2)
lines(11:15, preds + 1.96 * sqrt(diag(sigma_pred)), col = "red", lwd = 2, lty = 2)
lines(11:15, preds - 1.96 * sqrt(diag(sigma_pred)), col = "red", lwd = 2, lty = 2)
```


Tolerance
0.5
1.0
1.5
2.0
2.5
3.0
3.5
11
12
13
14
15
Age